

ENSEMBLE-AWARE STRATEGIES FOR TARGETING HIV-1 TAR

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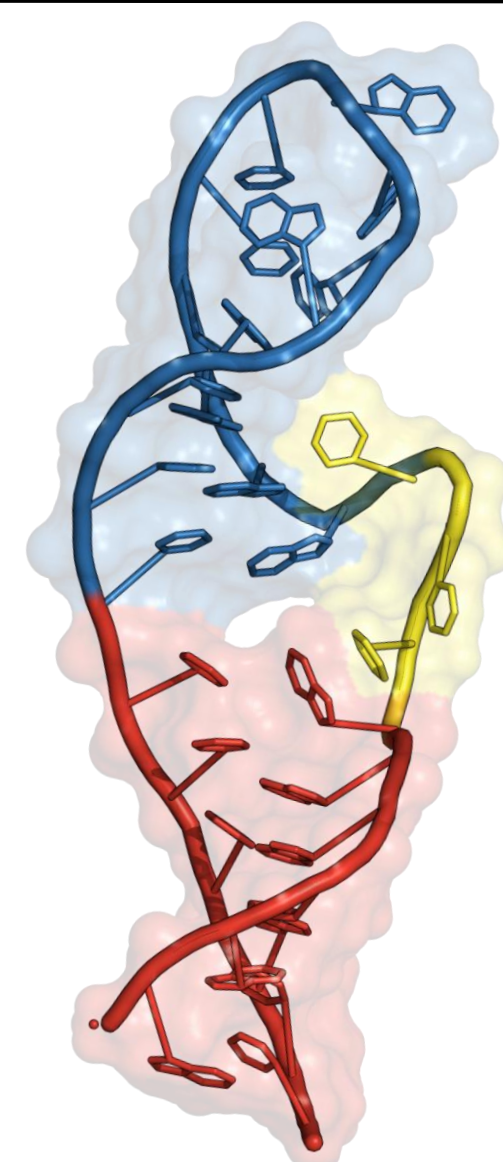
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Introduction

- The trans-activation response element (TAR) is an RNA stem-loop structure located at the 5' end of the HIV-1 genome [1].
- The conformation of the stem-loop is fundamental for the interaction with Integrase (IN) and Tat protein [2].
- No experimental structure of the TAR IN complex is currently available.



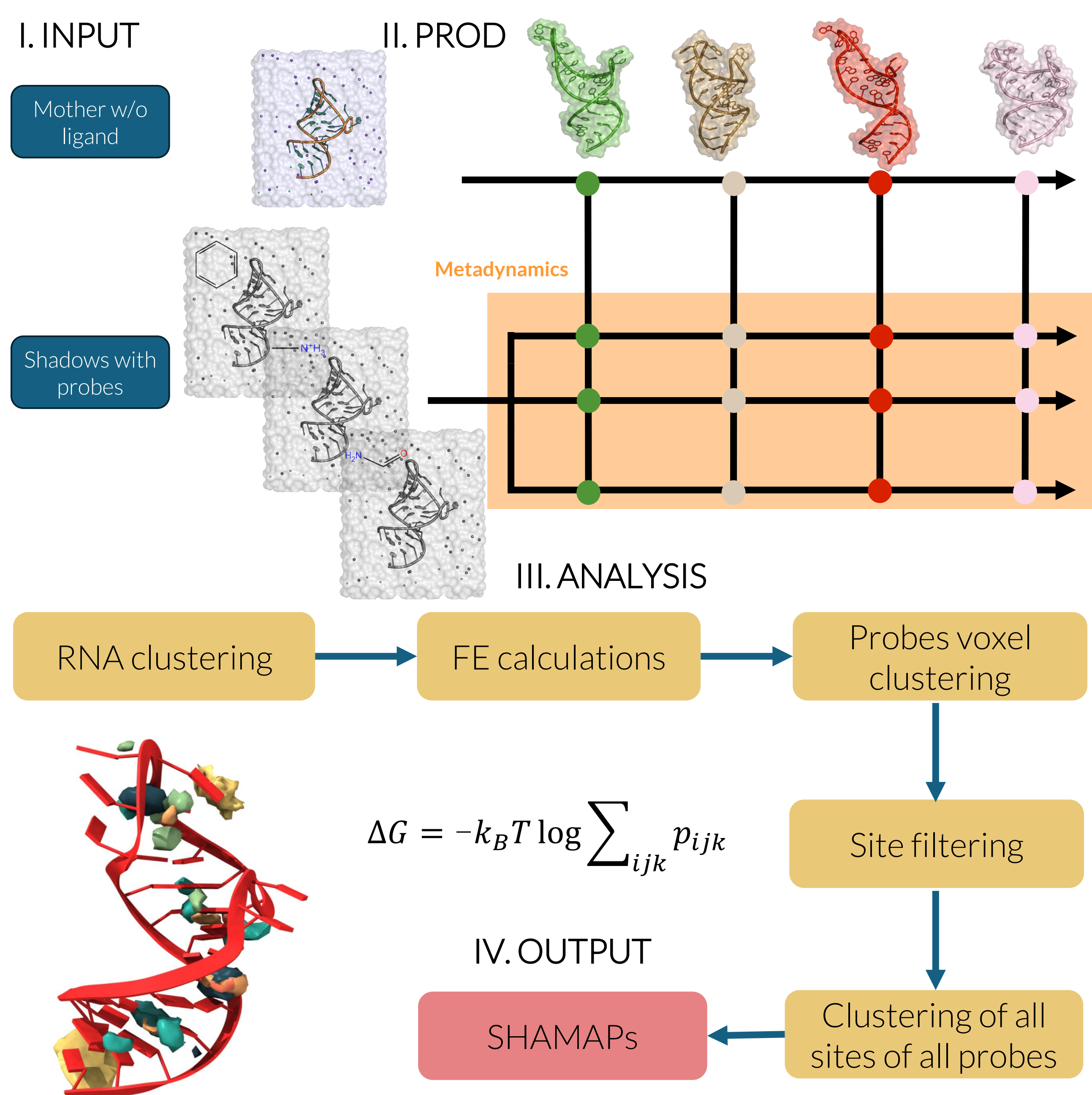
- Due to the high intrinsic flexibility of this small RNA, its conformational space is wide and complex.
- Residual dipolar couplings provide experimental information for ensemble level characterization.

1. DISCOVERY OF TAR-IN DISRUPTORS

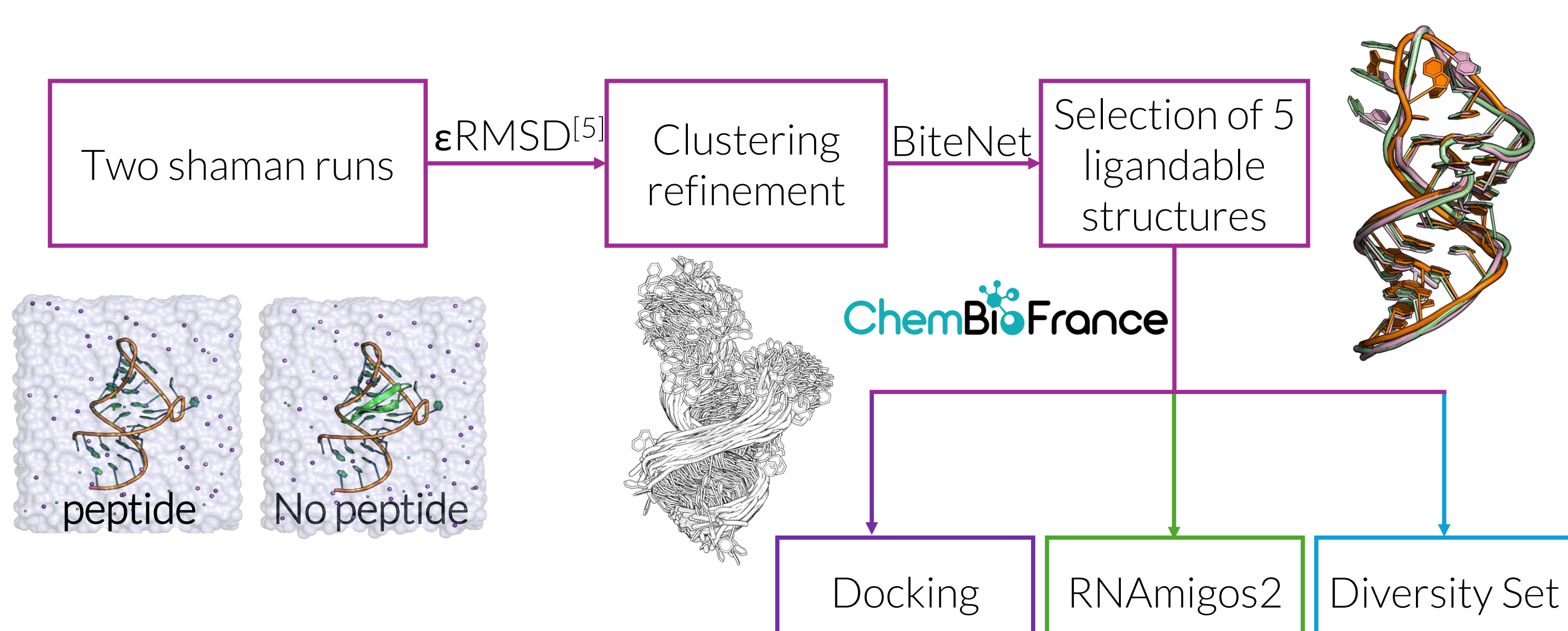
SHAMAN Protocol

SHAdow Mixed solvent metAdyNamics (SHAMAN) [3] is a computational tool for binding site identification in dynamic RNA structural ensembles.

- Exploration of **conformational landscapes** with molecular dynamics (MD) simulations.
- Identification of **small-molecule binding sites** by coupling mixed-solvent MD and metadynamics



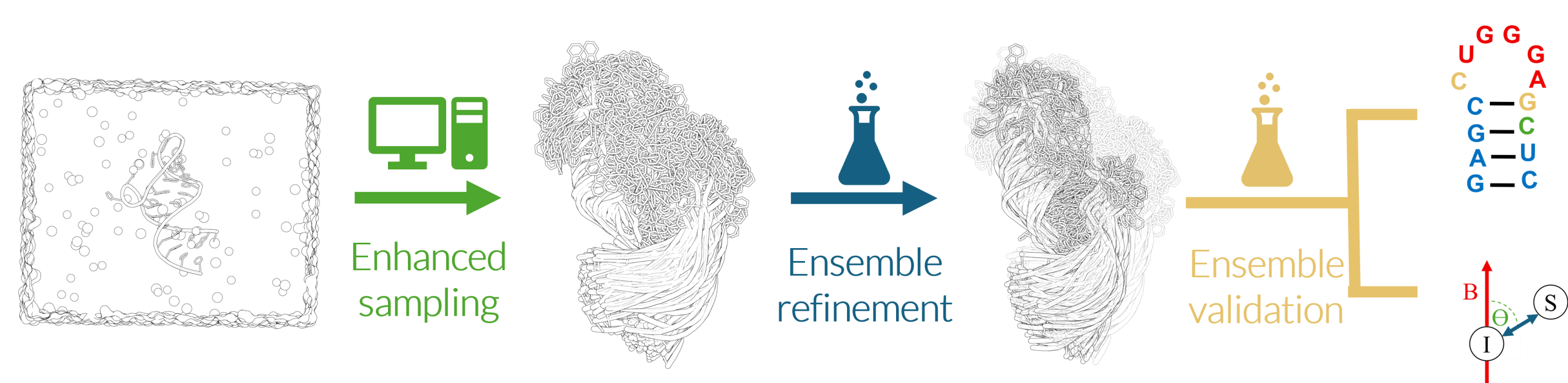
Virtual Screening Workflow



- 200 selected molecules tested as TAR-IN interaction disruptors.
- εRMSD as a metric improved cluster analysis within the SHAMAN workflow.

2. RECONSTRUCTION OF THE TAR CONFORMATIONAL ENSEMBLE

Integration Workflow



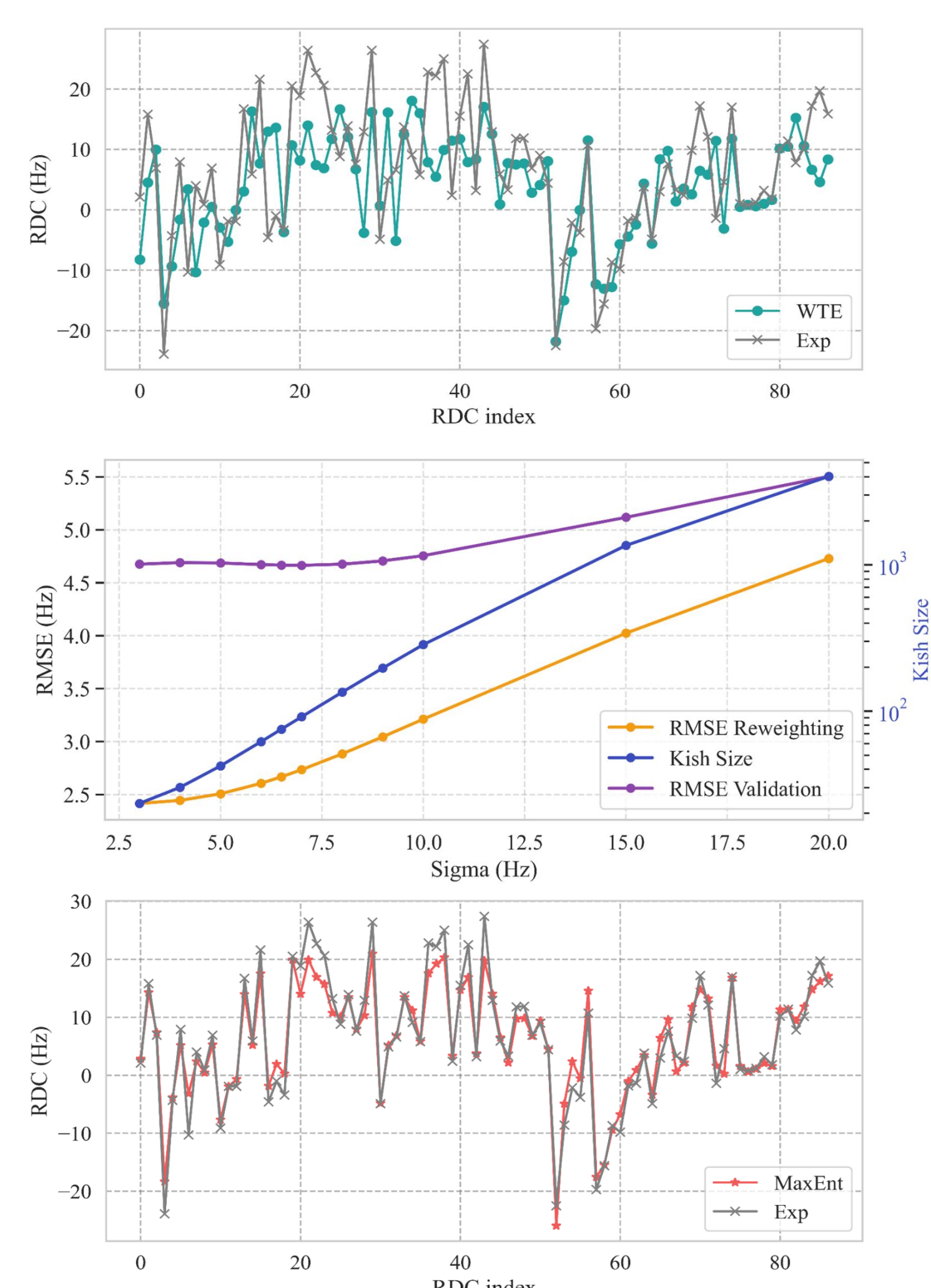
Maximum Entropy Reweighting

The Maximum Entropy reweighting [6] **minimally perturbs** the prior while correcting ensemble population weights.

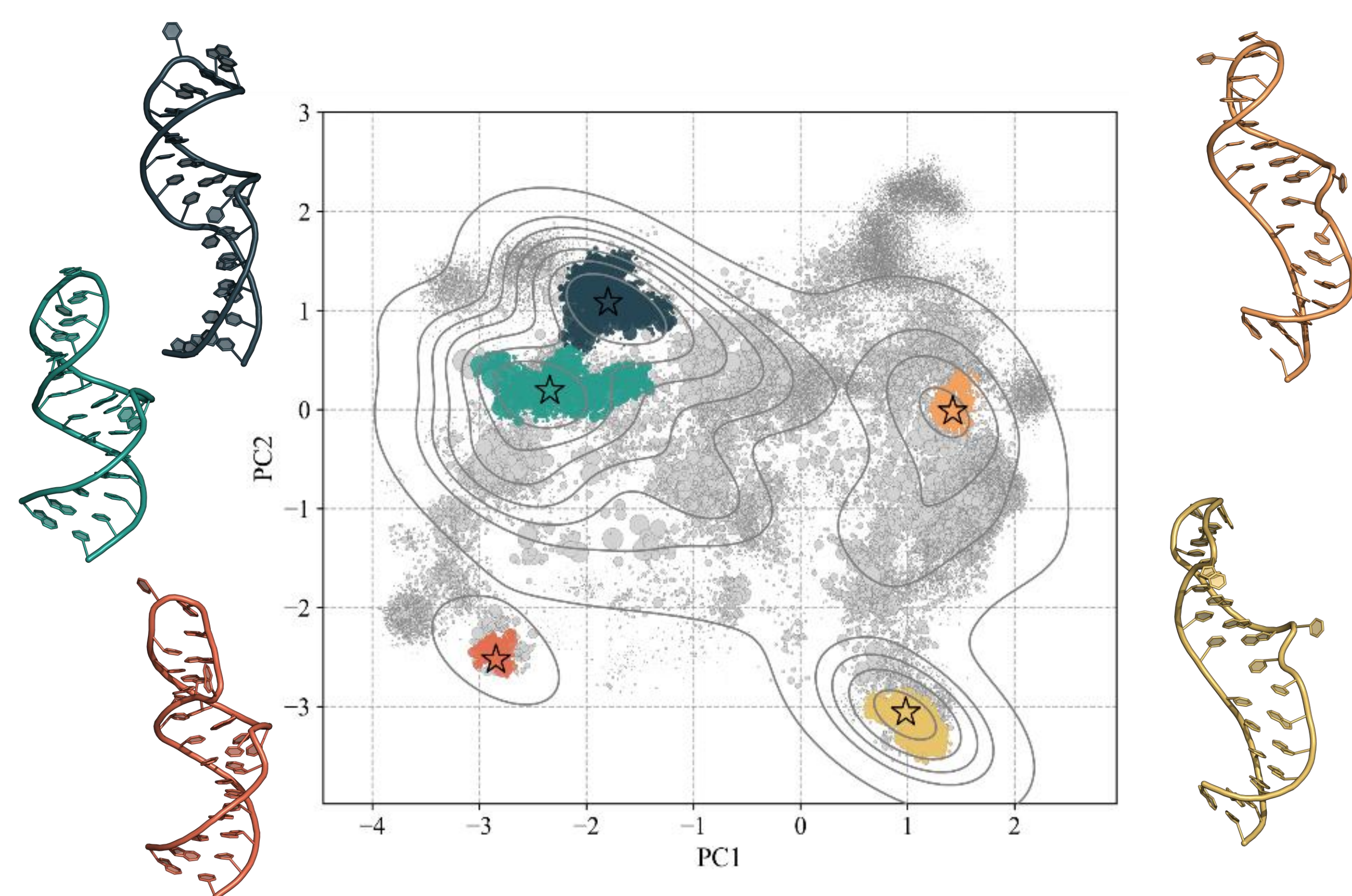
$$S[P||P_0] = - \int d\mathbf{q} P(\mathbf{q}) \ln \frac{P(\mathbf{q})}{P_0(\mathbf{q})}$$

$$\begin{cases} P_{ME}(\mathbf{q}) = \arg \max_{P(\mathbf{q})} S[P||P_0] \\ \int d\mathbf{q} s_i(\mathbf{q}) P(\mathbf{q}) = \langle s_i(\mathbf{q}) \rangle = s_i^{exp}, i = 1, \dots, M \\ \int d\mathbf{q} P(\mathbf{q}) = 1 \end{cases}$$

- Experimental noise modeled via a regularization term (σ)
- Improved RDC agreement
RMSE 8.4 Hz $\Rightarrow$ 2.8 Hz



Reconstructed Conformational Ensemble



- The integration of enhanced sampling simulations and experimental data yielded an **accurate conformational ensemble** of TAR.
- Representative cluster structures can be used as starting points for **multiple SHAMAN runs** to extend the explored conformational space.

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References

- [1] Chavali, S., et al. Journal of Biological Chemistry. 2019;294(24): 9326-9341.
[2] Rocchi C, et al. International Journal of Molecular Sciences. 2022;23(22):13742.
[3] Panei, F. P., et al. Nature Communications. 2024;15(1): 5725.

- [4] Bonomi, M., et al. Computer Physics Communications. 2009;180(10): 1961-1972
[5] Bottaro, S., et al. Nucleic Acids Research. 2014;42(21): 13306-13314
[6] Cesari, A et al. Computation. 2018;6(1), 15.

